Coursework Brief	
Module Code	CS34020 / CSM4220
Module Title	Computer Vision
Module Coordinator	Bernie Tiddeman (bpt@aber.ac.uk)
Member of Staff Setting the Coursework	Bernie Tiddeman (bpt@aber.ac.uk)
Coursework Title	Pick your own!
Weighting in module	50%
Submission Deadline	23rd November 2026
Submission information	***Presentation slides**: Please submit your slides to Blackboard.* ***Short report:** Please submit this to Turnitin.* ***Zip file of implementation**: Please submit a zip file of your software implementation to blackboard.* ***Participation:** in assessed group discussion.*
Word/length limit	***Presentation** of at most 8 content slides (recommended 5 slides), plus a title slide and summary slide (10 slides maximum).* ***Report** of up to 8 pages in length (12pt font such as Times New Roman or Arial, 1-inch (2.54 cm) margins, double-line or 1.5-line spacing, and left-aligned text for readability). Page length includes images (which should be included!).* *Penalties: 10% penalty will be applied for each page over the length limit. Similar penalties will be applied for incorrect formatting, particularly if correct formatting would result in exceeding the page limit.*

Learning Outcomes to be addressed by the coursework:

CS34020 and CSM4220:

Express a consolidated and extended understanding and knowledge of Computer Vision techniques.

Compare, critically evaluate and discuss competing methods.

Explain the problems, techniques and difficulties associated with the different areas of Computer Vision.

CSM4220 only:

Critically evaluate existing research in Computer Vision and demonstrate how this research may be applied appropriately to alternative application areas.

Written assignments MUST be in your own work. Do NOT take short cuts by copying even the smallest amount of material from anywhere (including the internet, books or another student's text). This is plagiarism and will be detected. N.B. Presenting work generated by AI as if it were your own is Unacceptable Academic Practice and you must not do this. AU procedures and penalties for 'Unacceptable Academic Practice' are strict and some may prevent you from qualifying for your degree see: https://www.aber.ac.uk/en/academic-registry/handbook/regulations/uap/

Coursework Submission

Submitting your assignment:

- How do I submit an assignment in Turnitin?
- How do I submit a Blackboard Assignment?

Academic integrity:

- Student guide to Referencing and Plagiarism
- Regulation on Unacceptable Academic Practice
- Guidance on Using Generative AI

Failure to follow the advice in the above links may result in a mark of **zero** if your submission is not uploaded, is in an unreadable format or fails to meet academic integrity requirements.

Clarification of coursework brief

It is your responsibility to ensure that you understand the requirements of the coursework that has been set. You should not expect a response in time to help if you ask for guidance less than three days before the deadline.

University policy on late/non-submission

Work submitted after the deadline without an approved extension will be given a mark of **zero**.

Extension to coursework deadlines

Extensions may be awarded in exceptional circumstances, e.g. on medical grounds, or for genuine, unforeseen personal circumstances such as illness, family problems or death of a relative or close personal friend. You may apply for an extension using the Coursework Extension Request Form. This should be submitted together with any supporting documentary evidence such as a medical certificate by email to the Faculty Registry Team (Extensions@aber.ac.uk). In order to be awarded an extension, you should submit the Extension Request Form (with appropriate evidence) at least **three working days before the deadline**. If insufficient evidence is submitted, then the extension may be awarded subject to further evidence being provided by a given date. More details are available here https://www.aber.ac.uk/en/academic-registry/handbook/taught-schemes/name-193261-en.html.

Failure of computers, printers and backups etc. will **not** be accepted as reasons for late submission of assessed work.
If an extension is granted you should follow the advice on submission provided by the Faculty of Sciences Extension team.
Please refer to your handbook for further information on Coursework Extensions.

If an extension has not been confirmed, then you should always submit an attempt of the work by the deadline.

Pick your own!

In this assignment you are going to develop a computer vision system intended to support a strawberry picking robot. No robotics is required, but the system should identify locations in images and videos, track the picking points in videos, and output "pick now" to the command line or screen to indicate that a strawberry had been identified and located for picking. The project will be broken down into several steps, and there will be an opportunity to extend to more complex additional rules. **Note, your solution should run on the CS lab machines without the need for additional libraries being installed.**

Part 1: Workshops (10%)

You are asked to submit the work done in each workshop, and to answer the question in the workshop handout. For CS34020 students this is generally reflecting on how the content of the workshop could (or couldn't) be used for the assignment. For CSM4220 students it will generally be how the contents of the workshop links to recent academic literature.

Part 2: Implementation (20%)

In the first step you are asked to decide on a suitable approach and develop the main program. The exact approach is up to you, and it would be good to see a variety of approaches (see Figure 1) in each discussion group (see later). Your implementation should use python OpenCV code to identify and track stem picking locations on strawberries. You are allowed to use a pre-trained model of your choice to do object detection, if you are using that kind of approach. In your report you will be asked to explain the choice of model, and to reflect on its efficiency and effectiveness in your report. You are asked **not** to train your own model, as it is too time consuming and processor intensive.

You have the choice of implementing the code yourself or with the use of AI. If you use AI, you should clearly describe the process used (the AI platform used, the prompts, skills, guardrails, etc.) in your report. If you implement it yourself, make this clear too. Implementing it yourself, you will probably have a deeper understanding and may find this helps later, e.g. in the discussion, but your implementation will probably be more basic as a result.

You should verify that your code is behaving correctly (as intended) and assess the quality of your approach using suitable testing and statistics.

Note that you do not get marks for code that you did not produce yourself, but a well-reasoned, evidenced and understood AI approach can obtain marks.



Figure 1 Some options for the vision system include semantic segmentation (left) or object detection (centre). The aim is to detect not just the strawberry, but also the picking points on the stem, which are about 1-2cm above the strawberry as shown on the right. Unripe strawberries should not be picked or damaged. Overripe or mouldy strawberries should be avoided if possible, but a second stage will discard them once picked if needed.

Part 3: Flair / optional extras (20%)

To achieve marks in this section you need to do something particularly impressive or well, going beyond the basic brief and showing an element of "flair". Note that you should have achieved all the basic functionality to some degree before attempting additional features. No marks will be awarded unless the basic functionality is mostly complete and working (otherwise, you have just implemented something different than what was required!). Flair could be a particularly well designed and engineered solution (explain why in your report), or there are several optional extensions to the basic system that you could attempt. For example, assessing the quality of the fruit (ripe, underripe, overripe, mouldy, etc), or the health of the plant (healthy, diseased, parched / wilting, yellowing, etc.), or could look at path planning for the picker, selecting the order / path to pick the strawberries.

Part 4: Report (20%)

Write a report on your implementation, with the following sections:

- **Introduction**: Give a brief overview of your solution and the main successes and limitations.
- **Implementation**: Explain your implementation and results (include example images, screenshots, code snippets / algorithms, etc)
- **Self-evaluation**: include a suggested mark (as a percentage) and your reason for each section and for the report itself.
- **References and statement on AI use**: Make it very clear (a) where you have got information or example code from, (b) if you have used AI or not to generate your solution or report, and (c) if you have got code or other resources from other sources (these must be referenced). **Note that you do not get marks for code that you did not produce yourself!**

Note that we will primarily be marking your reports and presentations and only checking or running your code to ensure that the report is accurate and truthful. Make sure all required information is in the report, including evidence and results (such as example images). If you haven't highlighted some feature in your report, don't be surprised if you don't get marks for it!

Part 5: Presentation and group discussion (30%)*

You will be assigned to small groups (e.g. 4-5 students) where you will be asked to present your solution, explain the pros and cons of the selected approach and how the correctness of the code has been verified and the quality has been evaluated. **Each presentation should take no more than 10 minutes.** After all members of the group have described their approach and results, the group will discuss the different implementations, and each member of the group should try to ascertain which of the presented methods would be most suitable for the intended application. You will be assessed on your engagement with the discussion, including both your questions and your answers. ***Note that the discussion will function partly as a check on your understanding of what you have done and the problem in general. Evidence of a lack of understanding will be accounted for when marking the other parts of the assignment.**

Several example videos are available on Blackboard. Some of these have been downloaded from YouTube (using yt-dlp: https://github.com/yt-dlp/yt-dlp) for your convenience. If you wish to view the originals, the YouTube ID is in square brackets in the file name, following the displayed name. Some have been down sampled and trimmed.

Assessment Criteria

You can access the assessment criteria (Turnitin Rubric) for your assignment by following the instructions in the following FAQ: https://faqs.aber.ac.uk/index.php?id=9271

A marking grid is given below.

	Fail	Third	2(ii)	2(i)	First
Part 1 (workshops 10%)	No attempt to answer questions from the workshops, or serious problems with the answers.	Some workshop answers submitted, some problems with the answers to the questions.	Good answers to most of the workshop questions.	Very good answers to the workshop questions.	Excellent answers to the workshop questions.
Part 2 (implementation 20%)	No attempt to detect strawberry picking locations or a very incomplete attempt or an attempt with very significant problems.	A reasonable attempt to use of OpenCV to detect strawberry picking locations, may be limited to static images / single frames, which has run into significant problems.	Reasonable use of OpenCV to detect strawberry picking locations with an attempt at tracking in videos, but has some problems, or may be incomplete.	Good use of OpenCV to detect strawberry picking locations in videos, with a good attempt at indicating when and which to pick.	Excellent use of OpenCV to detect strawberry picking locations in videos, with an excellent attempt at indicating when and which to pick.
Part 3 (flair/extras 20%)	No extra features / evidence of flair, or basic functionality not sufficiently complete to warrant extras.	Some simple additional features (e.g. additional colours) or somewhat thoughtful design or evaluation.	Some more ambitious additional features but incomplete or with problems, or thoughtful design or thorough evaluation.	Some more ambitious additional features complete, or very well thought out design or thorough evaluation.	Some really impressive additional features or excellent design or thorough evaluation.
Part 4 (report 20%)	Report missing or very incomplete. Very poor use of English, or missing figures or references, or missing self-evaluation, or very unclear descriptions of what has been done.	Report that is lacking in certain respects, such as poor use of English, poor use of figures, tables, referencing, results, or unclear descriptions of what has been done or poor self-evaluation.	Reasonable report, with reasonable English, use of figures, tables, referencing, results, and reasonably good descriptions of what has been done and reasonable self-evaluation.	Very good report, with very good English, use of figures, tables, referencing, results, and very good clear and concise descriptions of what has been done and very good self-evaluation.	Excellent report, with excellent English, use of figures, tables, referencing, results, and excellent clear and concise descriptions of what has been done and excellent self-evaluation.
Part 5 (presentation and discussion 30%)*	Non-participation in the presentation and discussion.	Participation in the discussion, but poor engagement or critical analysis.	Reasonable engagement in the discussion, with a decent set of slides and some critical analysis of the methods presented.	Good engagement in the discussion, with a good set of slides and critical analysis of the methods presented, showing strong understanding of the options presented.	Excellent engagement in the discussion, with an excellent set of slides and critical analysis of the methods presented, showing excellent understanding of the options presented.

***Note that the discussion will function partly as a check on your understanding of what you have done and the problem in general. Evidence of a lack of understanding will be accounted for when marking the other parts of the assignment.**